

DATA SCIENCE

The Data Science track develops graduates into industry-ready data professionals capable of extracting insights, building intelligent systems, and deploying AI solutions at scale. Across more than 1,455 hours of comprehensive training, participants acquire hands-on expertise in machine learning, deep learning, big data, cloud technologies, MLOps, and Generative AI, supported by strong foundation in programming, mathematics, statistics and business analytics. The track prepares graduate for high-demand careers in data science, AI engineering, machine learning, analytics and business intelligence across Egypt's banking, telecommunications, healthcare, retail and technology sectors.

JOB PROFILES

Data Scientist · Machine Learning Engineer · GenAI Engineer · MLOps Engineer · Data Analyst · Bigdata Analyst



TOTAL HOURS

1,455^{hrs}

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



CERTIFICATIONS

6

Industry-certified



Where graduates work



Certifications earned

-  **AI-901:** — Microsoft Certified: Azure AI Fundamentals
-  **AI-300:** — Microsoft Certified: Machine Learning Operations Engineer Associate
-  **AI-103:** — Microsoft Certified: Azure AI Apps and Agents Developer Associate
-  **DP-100:** — Microsoft Certified: Azure Data Scientist Associate
-  — AWS Certified AI Practitioner
-  — AWS Certified Machine Learning Engineer - Associate



Selection process

01 IQ & English Exam

-  Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

-  Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

-  Interview with the technical committee.

04 Interpersonal Interview

-  Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to all university graduates with a minimum grade of Good or higher.

Program content – 1,455 hours

CATEGORY	HOURS	SHARE	KEY COURSES
Fundamental Courses	144 hrs	10%	Database Fundamentals, Introduction to Gen AI & Prompt Engineering, Computer Networks Fundamentals, Operating Systems Fundamentals, Object Oriented Paradigm Using Java, Introduction to programming, Algorithms and Data Structure (Fundamentals),
Core Courses	954 hrs	66%	Financial Accounting, Introduction to Oracle SQL and PL/SQL, Oracle Advanced PL/SQL, Analytical SQL, Data Warehousing Fundamentals And Data Modeling, Bash Shell Scripting, Red Hat System Administration I, Business Statistics, Linear Algebra for Data Science, Calculus for Data Science, Numerical Optimization for Data Science, Data Analysis using Excel, Data Preparation & Preprocessing, Introduction to Modeling & Operations Research, Machine Learning, Data Scientist Toolkit, Advanced Machine Learning & Deep Learning, Time Series Analysis, SAS Programming Essentials using SAS Viya & Data Manipulation, Optimization and Simulation Methods for Analytics, Visualization & Storytelling, PowerBI for Data Science, Python For Data Science, Apache Hadoop Essentials, Apache Spark Essentials, Cloud Computing Fundamentals, Sequential modeling (including Transformers) and LLMs, Agentic AI (frameworks, architecture, evaluation), Build a chatbot using RAG (gain a solid understanding of RAG pipeline) (Workshop), Machine Learning Operations for Data Science (MLOps)
Graduation Project	300 hrs	21%	
Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Every graduate must obtain at least one freelancing job and an international certificate based on his track

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Database Fundamentals
<https://maharatech.gov.eg/course/view.php?id=740>
- Career Talk in Data Analytics
<https://maharatech.gov.eg/mod/hvp/view.php?id=4197>
- ITI Values & Mission
<https://iti.gov.eg/about-us>
- Good Fundamentals for Statistics, ML, AI, DL and understand the Data Analytics life cycle

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.